

Section two

1. Two dimensional stream function

- A. relates velocity and pressure.
- B. is constant along a stream line.
- C. is constant along an equipotential surface.
- D. none of these.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

2. What causes cavitation in centrifugal pump ?

- A. High suction pressure
- B. Low barometric pressure
- C. Low suction pressure
- D. High suction velocity

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

3. Draining of shallow pits or sump is done by a sump pump, which is a _____ pump.

- A. single stage vertical
- B. centrifugal
- C. plunger
- D. diffuser

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

4. A hydraulic ram acts as a/an _____ pump.

- A. centrifugal
- B. reciprocating
- C. impulse
- D. parallel cylinder

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

5. The ratio of average fluid velocity to the maximum velocity in case of laminar flow of a Newtonian fluid in a circular pipe is

- A. 0.5
- B. 1
- C. 2
- D. 0.66

Answer: Option A

6. Choose the correct set of dimensions of viscosity that are equivalent (where, F, M, L, T are dimensions for force, mass, length and time respectively).

- A. $FL^{-2}T$, $ML^{-1}T^{-1}$
- B. FLT , $ML^{-1}T^{-1}$
- C. $ML^{-1}T^{-3}$, $F^{-1}L^2T$
- D. $F^{-1}L^2T^{-1}$, MLT^{-3}

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

7. The line of action of the buoyant force passes through the centre of gravity of the

- A. submerged body.
- B. displaced volume of the fluid.
- C. volume of fluid vertically above the body.
- D. horizontal projection of the body.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

8. Pick out the wrong statement.

A.

The shear stress at the pipe (dia = D , length = L) wall in case of laminar flow of Newtonian fluids is

B.

In the equation,
the value of ' n ' for pseudoplastic and dilatant fluids are < 1 and > 1 respectively.

C.

Shear stress for Newtonian fluid is proportional to the rate of shear in the direction perpendicular to motion.

D.

With increase in the Mach number
>0.6, the drag co-efficient decreases
in case of compressible fluids.

Answer: Option D

Explanation:

No answer description available for this
question. Let us discuss .

[View Answer](#)

[Discuss](#)

9. The friction factor for turbulent flow in a
hydraulically smooth pipe

- A. depends only on Reynolds number.
- B. does not depend on Reynolds number.
- C. depends on the roughness.
- D. none of these.

Answer: Option A

Explanation:

No answer description available for this
question. Let us discuss .

[View Answer](#)

[Discuss](#)

10. The pressure at a point in a fluid is not the
same in all directions, when the fluid is
viscous and

A. moving

B. static

C. cold

D. hot

Answer: Option A

11. What causes convective acceleration in fluid flow ?

A. Steep slope in flow

B. Unsteady nature of flow

C. Non-uniformity of flow

D. Turbulence in flow

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

12. The Stoke's stream function applies to the

A. irrotational flow only.

B. ideal/non-viscous fluids only.

C. cases of axial symmetry.

D. none of these.

Answer: Option C

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

13. Slugging in a fluidised bed can be avoided by using

- A. tall narrow vessel.
- B. deep bed of solids.
- C. shallow beds of solids and proper choice of particle size.
- D. very large particles.

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

14. The ratio of width to depth for the most economical rectangular section in open channel flow is

- A. 0.5
- B. 1
- C. 1.5
- D. 2

Answer: Option D

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

15. The simple pitot tube measures the

_____ pressure.

A. static

B. dynamic

C. total

D. none of these

Answer: Option C

16. Capacity of a hydraulic accumulator is defined

in terms of maximum

A. amount of energy stored.

B. flow rate through accumulator.

C. rate of falling of ram.

D. volume available in the cylinder.

Answer: Option A

17. Three piping networks as shown in the figure are

placed horizontally. They are made using identical

pipe segments and are subjected to the same

pressure drop across them. Assuming no pressure

losses at junctions, the flow rates across the three

networks are related as $Q_1 : Q_2 : Q_3$.

A. 1: 3: 2

B. 1:2:3

C. 1:2:2

D. 1: 2: 2

Answer: Option C

Explanation:

No answer description available for this question.

Let us discuss .

[View Answer Discuss](#)

18. With diminishing cross-sectional area in case of subsonic flow in a converging nozzle, the

A. velocity increases

B. pressure decreases

C. both (a) & (b)

D. neither (a) nor (b)

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

19. Boundary layer separation is characterised by one of the conditions given below, where 'Re' is the Reynolds number for the flow. Select the appropriate conditions.

- A. $Re \ll 1$, accelerating flow
- B. $Re \gg 1$, accelerating flow
- C. $Re \ll 1$, decelerating flow
- D. $Re \gg 1$, decelerating flow

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

20. Manometers measure the _____ pressure.

- A. vacuum as well as the atmospheric
- B. difference in
- C. absolute
- D. gage

Answer: Option B

21. Which of the following fluid forces are not considered in the Navier-Stoke's equation?

- A. Gravity forces
- B. Viscous forces
- C. Pressure forces
- D. Turbulent forces

Answer: Option D

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

22. Which of the following may be termed as a variable orifice flowmeter ?

- A. Rotameter
- B. Pitot tube
- C. V-notch
- D. All (a), (b) and (c)

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

23. In turbulent flow, a rough pipe has the same friction factor as a smooth pipe

- A. in the zone of complete turbulence.
- B.

when the roughness projections are much smaller than the thickness of the laminar film.

- C. everywhere in the transition zone.
- D. when the friction factor is independent of the Reynold's number.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

24. Air vessel provided in a reciprocating pump

- A. smoothenes the flow by avoiding pulsations.
- B. increases the volumetric efficiency of the pump.
- C. saves the pump from the danger of cavitation.
- D. none of these.

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

25. Bernoulli's equation is dependent on the

- A. first law of thermodynamics.
- B. third law of thermodynamics.
- C. law of conservation of momentum.
- D. none of these.

Answer: Option D

26. The capacity of an accumulator is the maximum

- A. energy which it can store.
- B. discharge which it can deliver.
- C. liquid which it can store.
- D. none of these.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

27. With increase in the temperature, viscosity of a liquid

- A. increases
- B. decreases
- C. remains constant
- D. may increase or decrease; depends on the liquid

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

28. Vena-contracta pressure tapping is at a distance _____ from the position of an orificemeter fitted in a pipe of internal diameter 'd '

- A. d
- B. 0.5 d
- C. 2d
- D. 4d

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

29. The pump impeller and the turbine runner in a hydraulic torque converter

- A. have the same diameter.
- B. have different diameters.
- C. are directly coupled.
- D. none of these.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

30. Reciprocating pumps compared to centrifugal pumps

A. deliver liquid at uniform pressure.

B. can handle slurries more efficiently.

C. are not subject to air binding.

D. can be operated with delivery valve closed.

Answer: Option C

31. The fluid property which matters for fall-ing rain drops to acquire spherical shape is its

A. pressure

B. height of descend

C. viscosity

D. surface tension

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

32. The simple pitot tube does not measure the

A. static pressure.

B. dynamic pressure.

C. velocity at the stagnation point.

D. all (a), (b) and (c).

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

33. Boundary layer separation occurs when the

A. pressure reaches a minimum.

B. cross-section of the channel is reduced.

C. valve is closed in a pipeline.

D. velocity of sound is reached.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

34. The head loss in turbulent flow in a pipe varies

A. directly as the velocity.

B. inversely as the square of the velocity.

C. approximately as the square of the

velocity.

D. inversely as the square of the diameter.

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

35. For flow through an orifice from a reservoir, the actual velocity at the vena con-tracta is given by

A. $2gh$

B. $C_v \cdot 2gh$

C. $C_d \cdot 2gh$

D. $C_c \cdot 2gh$

Answer: Option B

36. The ratio of the hydraulic radius to the diameter of the channel, for maximum mean velocity of flow in a circular channel, in open channel flow is

A. 0.3

B. 0.9

C. 0.03

D. 0.66

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

37. For the free settling of a spherical particle through a fluid, the slope of, $C_D - \log N_{Re}$, plot is

- A. 1
- B. -1
- C. 0.5
- D. -0.5

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

38. A centrifugal pump loses prime after starting.

The reason of this trouble may be

- A. incomplete priming.
- B. too high a suction lift.
- C. low available NPSH and air leaks in the suction pipe.

D. all (a), (b), and (c).

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

39. Venturimeter and orificemeter measures the _____ of the fluid.

A. pressure

B. maximum velocity

C. average velocity

D. point velocity

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

40. Surge tanks are provided in high pressure water pipelines to

A. store a definite quantity of water all the time,

B. reduce the water hammer.

C. facilitate easy dismantling of pipeline

for cleaning and maintenance.

D. none of these.

Answer: Option B

41. Working principle of manometer comprises of balancing a column of liquid against the pressure to be measured. Inclined tube manometer is especially used for the measurement of _____ pressure.

A. small differential

B. atmospheric

C. absolute

D. gage

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

42. Isotropic turbulence occurs

A. where there is no velocity gradient.

B. at higher temperatures.

C. only in Newtonian fluids.

D. none of these.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

43. The co-efficient of drag and lift for an incompressible fluid depends on the

- A. Reynolds number
- B. Froude number
- C. Mach number
- D. all (a), (b) and (c)

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

44. Euler's equation of motion states, that at every point, the

- A. fluid momentum is constant.
- B. force per unit mass equals acceleration.
- C. rate of mass outflow is equal to the rate of mass inflow.
- D. none of these.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

45. The Navier-Stokes equation deals with the law of conservation of

- A. mass
- B. energy
- C. both (a) & (b)
- D. momentum

Answer: Option D

46. Hydraulic intensifier is used for increasing the

- A. rate of velocity of liquid supply.
- B. rate of flow through delivery pipeline of a pump.
- C. intensity of pressure of the liquid.
- D. momentum rate through delivery pipe.

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

47. Which of the following is the 'Blaussius equation',

relating friction factor to the Reynolds number ?

A. $f = 0.079 \cdot N Re^{-0.25}$

B. $f^{-0.5} = 4.07 \log_e (N Re f)^{-0.6}$

C.

D. none of these

Answer: Option A

Explanation:

No answer description available for this question.

Let us discuss .

[View Answer](#) [Discuss](#)

48. Hydrometer measures the specific gravity of liquids based on the principles of buoyancy.

Pycnometer is used to measure the specific gravity of

A. powder & granular solids

B. liquids

C. low melting point semi-solids

D. all 'a', 'b' & 'c'

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

49. Euler's equation of motion is a statement

expressing

- A. conservation of mass.
- B. conservation of energy.
- C. Newton's first law of motion.
- D. Newton's second law of motion.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

50. During ageing of fluid carrying pipes, the

- A. pipe becomes smoother with use.
- B. friction factor increases linearly with time.
- C. absolute roughness decreases with time.
- D. absolute roughness increases linearly with time.

Answer: Option D

.....end section two.....